

Project Design Phase-II

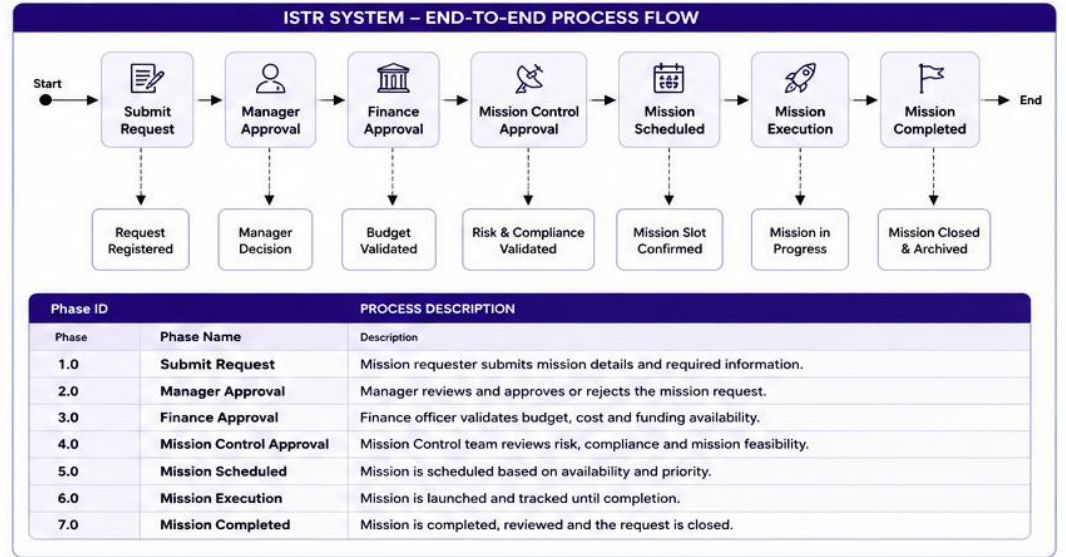
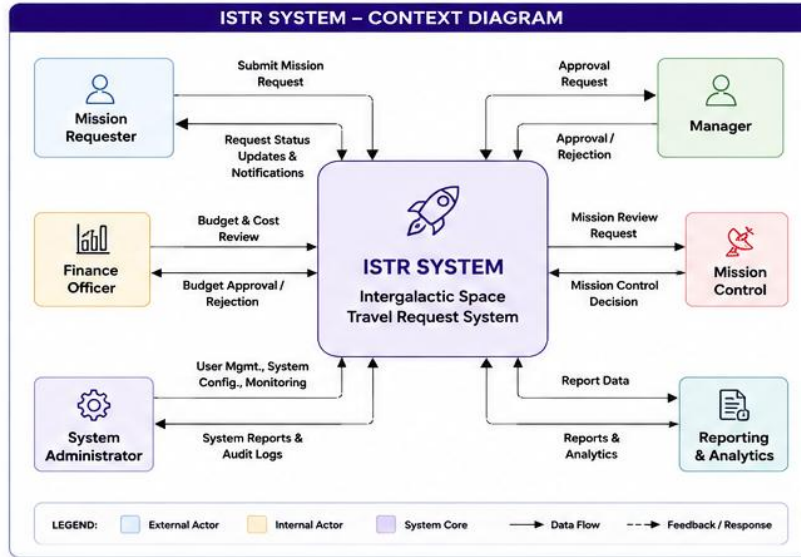
Data Flow Diagram & User Stories

Date	05 AUG 2026
Student Name	P. Srujan Sinha
Project Name	Intergalactic Space Travel Request (ISTR) System
Maximum Marks	4 Marks

Data Flow Diagrams (DFD)

A **Data Flow Diagram (DFD)** is a graphical representation that illustrates how data moves through a system, showing the flow of information between external entities, processes, data stores, and outputs. It helps stakeholders understand the system's functionality by clearly depicting how inputs are transformed into meaningful outputs. In the **Intergalactic Space Travel Request System (ISTR)**, the DFD demonstrates the complete lifecycle of a mission request—from submission by the requester, through manager, finance, and mission control approvals, to mission scheduling, execution, and completion.

The DFD is organized into different levels to provide increasing detail about the system. The **Level 0 (Context Diagram)** presents the ISTR system as a single process interacting with external actors such as the Mission Requester, Manager, Finance Officer, Mission Control, System Administrator, and Reporting & Analytics. The **Level 1 DFD** decomposes the system into major processes including request submission, approvals, budget validation, mission authorization, scheduling, and completion, while also showing the associated data stores. This structured representation improves system analysis, simplifies communication among stakeholders, and supports efficient system design, development, and maintenance.



User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Mission Requester	Mission Request Management	US-1	As a mission requester, I can submit a new mission request by filling in required details.	Mission request is submitted successfully and a unique request ID is generated.	High	Sprint 1
		US-2	As a mission requester, I can select supporting documents with my request.	Documents are uploaded successfully and linked to the request.	High	Sprint 1
		US-3	As a mission requester, I can track the status of my mission request.	Status is visible in real-time with accurate updates.	High	Sprint 1
		US-4	As a mission requester, I will receive email notifications for every update.	Email notifications are received for submission, approval and rejections.	Medium	Sprint 1
Manager	Approval Management	US-5	As a manager, I can review and take actions on mission requests assigned to me.	Manager can approve / reject the request.	High	Sprint 1
		US-6	As a manager, I can add comments while approving or rejecting a request.	Comments are saved and visible to the requester.	Medium	Sprint 1
Finance Officer	Budget Approval	US-7	As a finance officer, I can review budget details and validate funding.	Budget validation is updated with the approval status.	High	Sprint 2
		US-8	As a finance officer, I can request rejections with remarks if budget is not available.	Budget approval status is updated in the system.	High	Sprint 2
Mission Control	Mission Review & Authorization	US-9	As mission control, I can review mission feasibility and risk factors.	Mission check is complete and results are recorded.	High	Sprint 2
		US-10	As mission control, I can authorize the mission for scheduling.	Mission is authorized and status is updated to 'Scheduled'.	High	Sprint 2
System Administrator	System Administration	US-11	As an administrator, I can manage users, roles and permissions.	Users and roles are created and updated successfully.	High	Sprint 1
		US-12	As an administrator, I can manage catalogs, forms and approval workflows.	Catalogs and workflows are created and activated successfully.	High	Sprint 1